UNIT-2

1. Guided Transmission Media

Guided Transmission Media refers to communication channels in which data signals travel through a **physical path (cables or wires)** from the sender to the receiver. The signals are guided along a fixed route, which provides secure, reliable, and high-speed communication.

It is also known as **Bounded Media** or **Wired Communication Media**.

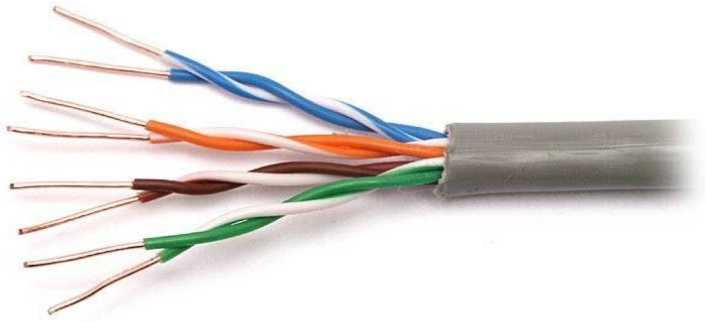
Types of Guided Transmission Media

There are **three main types**:

1. Twisted Pair Cable
2. Coaxial Cable
3. Fiber Optic Cable

1. Twisted Pair Cable

A **Twisted Pair Cable** is the most commonly used **guided transmission medium** in computer networks and telephone systems. It consists of **two insulated copper wires twisted together** to reduce electromagnetic interference (EMI) and crosstalk.



A twisted pair cable consists of:

1. **Copper Conductors**
   - Two thin copper wires carry electrical signals.
2. **Insulating Material**
   - Each wire is covered with plastic insulation to prevent short circuits.
3. **Twisting**
   - The wires are twisted together at regular intervals to reduce interference.
4. **Outer Jacket**
   - A plastic covering protects the cable from physical damage.

Working Principle

1. The sender converts data into **electrical signals**.
2. The electrical signals travel through the twisted copper wires.
3. Twisting reduces electromagnetic interference (EMI) and crosstalk.
4. The receiver converts the electrical signals back into data.

Types of Twisted Pair Cable

There are **two main types**:

1. Unshielded Twisted Pair (UTP)

UTP cable has **no metallic shielding** around the twisted wires. It is the most commonly used cable for Ethernet networks.

Applications

- Ethernet LAN
- Telephone lines
- DSL Internet

2. Shielded Twisted Pair (STP)

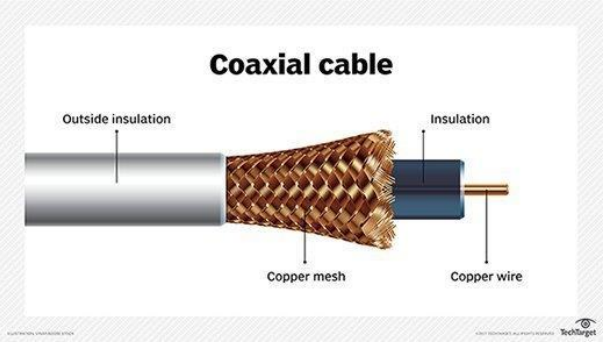
STP cable contains a **metallic shield (foil or braided mesh)** around the twisted wires to protect against electromagnetic interference.

Applications

- Industrial networks
- Hospitals
- Factories

2. Coaxial Cable

A **Coaxial Cable** is a type of **guided transmission medium** used to transmit data, voice, and video signals. It consists of a **central copper conductor** surrounded by insulation, a metallic shield, and an outer protective jacket. The shielding protects the signal from external electromagnetic interference (EMI), making coaxial cables more reliable than twisted pair cables.



A coaxial cable consists of **four main layers**:

1. Central Copper Conductor

- Located at the center of the cable.
- Carries the electrical signals (data).

2. Dielectric Insulator

- Surrounds the central conductor.
- Prevents signal leakage.
- Maintains the distance between the conductor and the shield.

3. Metallic Shield

- Made of braided copper or aluminum foil.
- Protects the signal from electromagnetic interference (EMI) and radio frequency interference (RFI).

4. Outer Plastic Jacket

- Covers the entire cable.
- Protects it from moisture, heat, and physical damage.

Working Principle

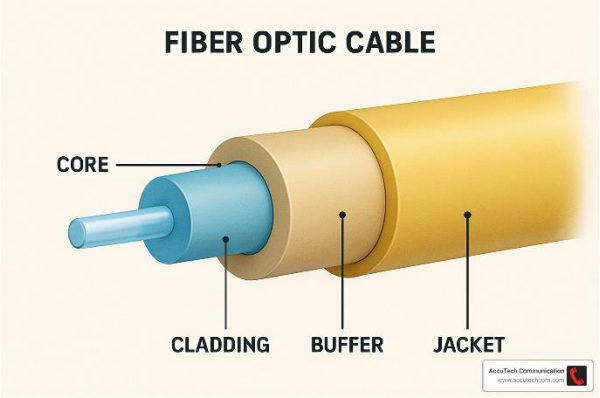
1. The sender converts data into **electrical signals**.
2. The signals travel through the **central copper conductor**.
3. The **dielectric insulator** prevents signal leakage.
4. The **metallic shield** blocks external noise and interference.
5. The receiver receives the signal with minimal loss.

Applications of Coaxial Cable

- Cable Television (CATV)
- Broadband Internet connections
- CCTV surveillance systems
- Cable modems

3. Fiber Optic Cable

A **Fiber Optic Cable** is one of the most advanced **guided transmission media** used in modern communication systems. Unlike copper cables, it transmits data in the form of **light pulses** through thin strands of glass or plastic fibers. It offers **very high speed, large bandwidth, long-distance communication, and excellent security**, making it the preferred medium for high-speed Internet, telecommunication, and networking.



A fiber optic cable consists of **four main parts**.

1. Core

- Central part of the cable.
- Made of glass or plastic.
- Carries light signals.

2. Cladding

- Surrounds the core.
- Has a lower refractive index than the core.
- Causes Total Internal Reflection.
- Prevents light from escaping.

3. Buffer Coating

- Protective plastic layer.
- Protects the fiber from moisture and physical damage.

4. Outer Jacket

- Outermost protective covering.
- Protects against heat, chemicals, and mechanical damage.

Working Principle

The working of a fiber optic cable involves the following steps:

1. The sender converts electrical data into **light pulses** using a **LED or Laser**.
2. The light pulses enter the **core** of the fiber.
3. The light travels through the core by **Total Internal Reflection (TIR)**.
4. The **cladding** keeps the light confined inside the core.
5. At the receiver, a **photodetector** converts the light pulses back into electrical signals.
6. The electrical signals are then processed as data.

Applications of Fiber Optic Cable

- High-speed Internet (FTTH)
- Telephone communication
- Cable television
- Data centers
- Computer networks

2. Functions of Data Link Layer

The **Data Link Layer** is the **second layer** of the **OSI (Open Systems Interconnection) Reference Model**. It lies between the **Physical Layer** and the **Network Layer**. The main responsibility of the Data Link Layer is to ensure **reliable, error-free, and efficient communication** between two directly connected devices. It converts the raw bit stream received from the Physical Layer into **frames**, detects and corrects errors, controls the flow of data, and manages access to the communication channel.

The major functions of the Data Link Layer are:

1. Framing
2. Physical Addressing
3. Error Control
4. Flow Control
5. Medium Access Control (MAC)
6. Reliable Delivery

1. Framing

Framing is the process of dividing a continuous stream of bits into **small units called frames** for easy transmission and error detection.

Purpose

- Organizes data into manageable units.
- Makes error detection easier.
- Identifies the beginning and end of data.

2. Physical Addressing (MAC Addressing)

The Data Link Layer adds the **Physical Address (MAC Address)** of the sender and receiver to each frame.

Purpose

- Identifies devices in the local network.
- Ensures the frame reaches the correct destination.

3. Error Control

Error Control ensures that transmitted frames are received **without errors**.

Functions

- Detects transmission errors.
- Corrects errors.
- Retransmits damaged or lost frames.

4. Flow Control

Flow Control ensures that the sender transmits data at a speed that the receiver can handle.

Need

If the sender sends data too fast, the receiver's buffer may overflow, causing data loss.

5. Medium Access Control (MAC)

Medium Access Control determines **which device can transmit data** when multiple devices share the same communication channel.

Purpose

- Avoids collisions.
- Provides fair access.
- Improves network performance.

6. Reliable Delivery

Reliable Delivery ensures that **every frame reaches the destination correctly and in the correct order**.

How It Works

- Uses acknowledgments (ACK).
- Retransmits lost or damaged frames.
- Maintains sequence numbers.

3. Data Link Layer Design Issues

The **Data Link Layer** is the **second layer** of the **OSI (Open Systems Interconnection) Reference Model**. It is responsible for providing **reliable and error-free communication** between two directly connected devices over a physical communication link. It converts the raw bit stream received from the Physical Layer into **frames** and ensures that these frames are delivered correctly to the destination.

The major design issues are:

1. Providing Services to the Network Layer
2. Framing

3. Error Control
4. Flow Control
5. Medium Access Control (MAC)

1. Providing Services to the Network Layer

The Data Link Layer provides services to the **Network Layer** by ensuring that data is delivered correctly between adjacent nodes.

Types of Services

a) Unacknowledged Connectionless Service

- No connection is established.
- No acknowledgment (ACK) is sent.
- Lost frames are not retransmitted.
- Fast but less reliable.

b) Acknowledged Connectionless Service

- No connection setup.
- Each frame is acknowledged.
- Lost frames are retransmitted.
- More reliable than connectionless service.

c) Acknowledged Connection-Oriented Service

- Connection is established before communication.
- Frames are delivered in sequence.
- Every frame is acknowledged.
- Connection is terminated after transmission.

2. Framing

Framing is the process of dividing a continuous stream of bits into **small units called frames**.

Each frame contains:

- Header
- Data
- Trailer

Methods of Framing:

a) Character Count

- The first field specifies the frame length.
- Simple but errors in the length field can affect synchronization.

b) Byte Stuffing

- Uses special flag bytes to mark the beginning and end of a frame.
- If a flag appears in the data, an escape byte is inserted before it.

c) Bit Stuffing

- Uses a special flag pattern (01111110).
- Whenever five consecutive 1s occur in the data, the sender inserts a 0.
- The receiver removes the extra 0 during reception.

Advantages of Framing

- Organizes data into manageable units.
- Simplifies error detection.
- Improves synchronization between sender and receiver.

3. Error Control

Error control ensures that transmitted frames reach the receiver **without errors**.

Functions

- Detect transmission errors.
- Correct errors.
- Retransmit damaged or lost frames.

Techniques

- Parity Bit
- Checksum
- Cyclic Redundancy Check (CRC)
- Hamming Code

Advantages

- Reliable communication.
- Reduces data corruption.
- Ensures data integrity.

4. Flow Control

Flow control ensures that the **sender does not transmit data faster than the receiver can process**.

Without flow control, frames may be lost because the receiver's buffer becomes full.

Methods:

a) Stop-and-Wait Flow Control

- The sender sends one frame.
- Waits for an acknowledgment (ACK).
- Sends the next frame only after receiving the ACK.

Advantages

- Simple to implement.
- Reliable.

b) Sliding Window Flow Control

- The sender can send multiple frames before receiving acknowledgments.
- Improves channel utilization.
- Increases transmission speed.

Advantages

- High efficiency.
- Better bandwidth utilization.

6.Medium Access Control (MAC)

When multiple devices share the same communication channel, **Medium Access Control (MAC)** determines **which device is allowed to transmit data** at a given time.

Purpose

- Prevent collisions.
- Ensure fair channel access.
- Improve network efficiency.

4. Framing & its Methods

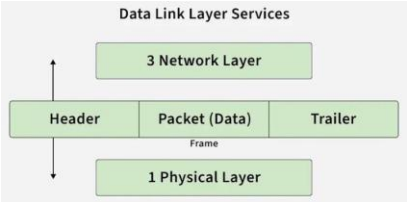
Framing is the process of dividing a continuous stream of bits into **small blocks called frames** by adding a **header** and **trailer** to the data. These frames are then transmitted over the communication channel.

Need for Framing

Framing is required because:

- A continuous stream of bits is difficult to identify.
- It helps the receiver recognize the **beginning and end** of each message.
- It makes **error detection and correction** easier.

A frame consists of three main parts:



Header

- Contains source and destination addresses.
- Contains control information.

Data

- Contains the actual information to be transmitted.

Trailer

- Contains error detection information (CRC or checksum).

Working of Framing

The framing process works as follows:

1. The Network Layer sends a continuous stream of bits.
2. The Data Link Layer divides the stream into frames.
3. A header and trailer are added to each frame.
4. Frames are transmitted through the Physical Layer.
5. The receiver identifies frame boundaries.
6. Error checking is performed before delivering the data to the Network Layer.

Methods of Framing

There are **three methods of framing**:

1. Character Count
2. Byte Stuffing
3. Bit Stuffing

1. Character Count Method

In this method, the **first field of the frame specifies the total number of bytes** present in the frame.

Working

- Sender counts the total number of bytes.
- The count is placed at the beginning of the frame.

- Receiver reads the count and knows where the frame ends.

Example

A parcel has a label stating **"8 items inside."** The receiver knows exactly how many items to expect.

2. Byte Stuffing (Flag Byte Method)

In this method, **special flag bytes** are placed at the beginning and end of every frame. If the same flag appears inside the data, an **escape byte (ESC)** is inserted before it.

Working

1. Sender places FLAG at both ends.
2. If FLAG appears in data, ESC is inserted.
3. Receiver removes ESC and recognizes the actual data

3. Bit Stuffing

In bit stuffing, a **special bit pattern (01111110)** is used as the frame delimiter. Whenever the sender encounters **five consecutive 1s** in the data, it automatically inserts a **0**.

Working

1. Sender inserts a 0 after every five consecutive 1s.
2. Receiver removes the extra 0.
3. The flag pattern is never confused with data.

Example

Imagine placing a **special separator** after every five identical books on a shelf so that counting becomes easier.

5. Error Control

In a computer network, data is transmitted from the sender to the receiver through a communication channel. During transmission, data may become corrupted due to **noise, interference, attenuation, or hardware failures**. The **Data Link Layer** uses **Error Control** techniques to detect and correct these errors, ensuring that the receiver gets the correct data.

Error Control is the process of **detecting, correcting, and retransmitting corrupted or lost data frames** to ensure accurate and reliable communication between the sender and receiver.

Techniques Used for Error Control

Error Control consists of two major processes:

1. Error Detection
2. Error Correction

1. Error Detection

Error Detection identifies whether the received data contains any errors.

Common Error Detection Techniques:

a) Parity Bit

A **Parity Bit** is the **simplest error detection technique** used in computer networks. It is an **extra bit (0 or 1)** added to the original data before transmission to check whether the data has been transmitted correctly.

Need for Parity Bit

During data transmission, bits may change because of:

- Noise
- Electromagnetic interference
- Hardware faults
- Signal attenuation

To detect these errors, an extra bit called the **parity bit** is added.

Working of Parity Bit

1. The sender counts the number of **1's** in the data.
2. Based on the parity type (Even or Odd), it adds one extra bit (0 or 1).
3. The receiver again counts the number of **1's**.
4. If the parity does not match, an error is detected.

Types of Parity Bit

There are **two types**:

1. Even Parity
2. Odd Parity

1. Even Parity

In **Even Parity**, the total number of **1's** (including the parity bit) must always be **even**.

Example

Original Data-1101000

Number of 1's = **3 (Odd)**

To make it **even**, add parity bit **1**.

Transmitted Data-11010001

Now total number of 1's = **4 (Even)**

2. Odd Parity

In **Odd Parity**, the total number of **1's** (including the parity bit) must always be **odd**.

Example

Original Data-1011001

Number of 1's = **4 (Even)**

To make it **odd**, add parity bit **1**.

Transmitted Data-10110011

Now total number of 1's = **5 (Odd)**

b) Checksum

A checksum is calculated by adding data segments together.

Working

1. Sender calculates checksum.
2. Appends checksum to data.
3. Receiver recalculates checksum.
4. If both match, the data is accepted.

c) Cyclic Redundancy Check (CRC)

CRC is the most widely used error detection technique. Is calculated by binary division added data segments together

Working

1. Sender calculates CRC value.
2. Appends CRC to data.
3. Receiver recalculates CRC.
4. If both match, the data is accepted.

2. Error Correction

Error Correction not only detects errors but also **corrects them** without requiring retransmission (or with retransmission when necessary).

When data is transmitted from one device to another, noise or interference may change some bits. For example:

Sent data: `1011001`
Received data: `1010001`

Here, one bit has changed from `1` to `0`. Error correction techniques help the receiver find and correct such errors.

Error correction generally follows these steps:

Original Data → Add Redundant Bits → Transmit → Receive → Detect Error → Correct Error

Common Error-Correction Techniques

1. Hamming Code

Hamming code adds parity bits to the data. It can identify the position of a single-bit error and correct it.

2. Reed-Solomon Code

Reed-Solomon code is effective for correcting **multiple errors and burst errors**. It is commonly used in CDs, DVDs, QR codes, and digital communication.

3. BCH Code

BCH (Bose–Chaudhuri–Hocquenghem) code can detect and correct multiple-bit errors and is used in various storage and communication systems.

6. Error-Correcting Codes

During data communication, the transmitted data may be affected by **noise and interference**. As a result, some bits may change from **0 to 1** or from **1 to 0**. This causes errors in the received data. To detect and correct these errors, **error-correcting codes** are used.

An **error-correcting code (ECC)** is a technique in which extra or redundant bits are added to the original data before transmission. The receiver uses these additional bits to identify the presence of errors and, in many cases, correct them without asking the sender to retransmit the data.

Errors can occur due to:

- Electrical or electromagnetic interference
- Noise in the communication channel
- Weak signals

The basic process of error correction is:

Original Data → Encoder → Codeword → Communication Channel → Decoder → Corrected Data

1. The sender provides the original data to the **encoder**.
2. The encoder adds redundant bits to the data.
3. The resulting data is called a **codeword**.
4. The codeword is transmitted through the communication channel.
5. Noise may change one or more bits.
6. The receiver receives the possibly corrupted codeword.
7. The **decoder** analyzes the received codeword using the redundant bits.
8. The decoder detects and corrects the error and produces the original data.

Types of Error-Correcting Codes

Some important error-correcting codes are:

1. Hamming Code

Hamming code is a commonly used error-correcting code that can **detect and correct single-bit errors**.

It adds parity bits at specific positions in the data. The receiver checks these parity bits to determine whether an error has occurred and identifies the position of the erroneous bit.

For example, if one bit changes during transmission, Hamming code can identify its position and correct it.

Applications: Computer memory, digital communication, and data transmission systems.

2. Reed-Solomon Code

Reed-Solomon code is a powerful error-correcting code that works particularly well with **burst errors**, where several consecutive bits or symbols are affected.

It adds redundant symbols to the original data. The receiver uses these symbols to reconstruct the original data even when some transmitted symbols are corrupted.

Applications: CDs, DVDs, QR codes, satellite communication, and digital television.

3. BCH Code

BCH (Bose–Chaudhuri–Hocquenghem) code is a type of cyclic error-correcting code that can be designed to correct multiple-bit errors.

It provides flexible error-correction capabilities and is widely used in digital storage and communication systems.

Applications: Flash memory, storage devices, and digital communication.

7. Error-Detecting Codes

During data communication, data may be affected by **noise, interference, signal distortion, or hardware problems**. These problems can change one or more bits of the transmitted data. To ensure reliable communication, the receiver must determine whether the received data contains an error.

An **error-detecting code** is a technique used to **detect whether an error has occurred during data transmission**. In this method, some extra bits called **redundant bits** are added to the original data before transmission. The receiver uses these bits to check whether the received data is correct or corrupted.

The basic process of error detection is:

Original Data → Add Redundant Bits → Transmit → Receive → Check Data → Detect Error

1. The sender takes the original data.
2. Additional redundant bits are added to the data.
3. The resulting data is transmitted through the communication channel.
4. Noise may change one or more bits during transmission.
5. The receiver checks the received data using the redundant bits.
6. If the checking condition is not satisfied, the receiver concludes that an error has occurred.

Common Error Detection Techniques:

a) Parity Bit

A **Parity Bit** is the **simplest error detection technique** used in computer networks. It is an **extra bit (0 or 1)** added to the original data before transmission to check whether the data has been transmitted correctly.

Need for Parity Bit

During data transmission, bits may change because of:

- Noise
- Electromagnetic interference
- Hardware faults
- Signal attenuation

To detect these errors, an extra bit called the **parity bit** is added.

Working of Parity Bit

5. The sender counts the number of **1's** in the data.
6. Based on the parity type (Even or Odd), it adds one extra bit (0 or 1).

7. The receiver again counts the number of **1's**.
8. If the parity does not match, an error is detected.

Types of Parity Bit

There are **two types**:

3. Even Parity
4. Odd Parity

1. Even Parity

In **Even Parity**, the total number of **1's** (including the parity bit) must always be **even**.

Example

Original Data-1101000

Number of 1's = **3 (Odd)**

To make it **even**, add parity bit **1**.

Transmitted Data-11010001

Now total number of 1's = **4 (Even)**

2. Odd Parity

In **Odd Parity**, the total number of **1's** (including the parity bit) must always be **odd**.

Example

Original Data-1011001

Number of 1's = **4 (Even)**

To make it **odd**, add parity bit **1**.

Transmitted Data-10110011

Now total number of 1's = **5 (Odd)**

b) Checksum

A checksum is calculated by adding data segments together.

Working

5. Sender calculates checksum.

6. Appends checksum to data.
7. Receiver recalculates checksum.
8. If both match, the data is accepted.

c) Cyclic Redundancy Check (CRC)

CRC is the most widely used error detection technique. Is calculated by binary division added data segments together

Working

5. Sender calculates CRC value.
6. Appends CRC to data.
7. Receiver recalculates CRC.
8. If both match, the data is accepted.

8. Elementary Data Link Protocols

The **Data Link Layer** is responsible for providing reliable communication between two directly connected devices. It receives packets from the Network Layer, organizes them into **frames**, and delivers them to the destination device.

To control the exchange of frames between the sender and receiver, different **data link protocols** are used. The basic protocols designed to explain the operation of the Data Link Layer are called **Elementary Data Link Protocols**.

These protocols mainly deal with **framing, flow control, error control, and reliable delivery of frames**.

Types of Elementary Data Link Protocols

The important elementary data link protocols are:

1. Unrestricted Simplex Protocol
2. Simplex Stop-and-Wait Protocol
3. Simplex Protocol for a Noisy Channel

1. Unrestricted Simplex Protocol

The **Unrestricted Simplex Protocol** is the simplest data link protocol.

Working:

- Data transmission takes place in **one direction only**.
- The sender always has data to transmit.
- The receiver is assumed to be always ready to receive data.
- The communication channel is assumed to be error-free.
- There is no need for acknowledgements.
- There is no flow control.

Sender **Receiver**

| |

|-------- **Frame 1** -------->|

|-------- **Frame 2** -------->|

|-------- **Frame 3** -------->|

2. Simplex Stop-and-Wait Protocol

The **Simplex Stop-and-Wait Protocol** is designed to handle the situation where the sender may transmit data faster than the receiver can process it.

Here, **flow control** is used.

The sender sends one frame and then **waits for the receiver to accept it** before sending the next frame.

Working:

1. The sender receives a packet from the Network Layer.
2. It creates a frame.
3. The frame is transmitted to the receiver.
4. The sender waits until the receiver is ready for the next frame.
5. The receiver accepts the frame and processes it.
6. The sender then sends the next frame.

Sender **Receiver**

| |

|-------- **Frame 1** -------->|

| |

|-------- **Frame 2** -------->|

| |

|-------- **Frame 3** -------->|

3. Simplex Protocol for a Noisy Channel

The **Simplex Protocol for a Noisy Channel** is designed for communication in which frames may be **lost or damaged**.

Unlike the previous protocols, this protocol considers transmission errors.

Working

- The sender sends a frame and waits for an acknowledgement from the receiver.
- If the receiver receives the frame correctly, it sends an **ACK (Acknowledgement)**.
- If the frame is damaged or lost, the sender does not receive the expected acknowledgement and retransmits the frame.

Sender Receiver

| |

|-------- **Frame 1** -------->|

|<---------- **ACK** ---------|

| |

|-------- **Frame 2** -------->|

|<---------- **ACK** ---------|

9. Initial Simplifying Assumptions and Basic Transmit and Receipt

The **initial simplifying assumptions** provide a simple environment for understanding Data Link Layer communication. The basic transmit operation takes a packet from the Network Layer, converts it into a frame, and sends it through the communication channel. The basic receive operation accepts the frame, extracts the packet, and delivers it to the Network Layer.

Initial Simplifying Assumptions

Before studying elementary data link protocols, the following assumptions are commonly made.

1. Two Devices are Communicating

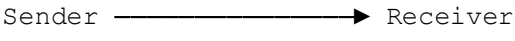
It is assumed that there are only **two devices** involved in communication:

Sender → Receiver

The sender transmits frames, and the receiver accepts them.

2. Communication is Simplex

Initially, communication is assumed to take place in **one direction only**.



The sender sends data, while the receiver only receives data.

3. Error-Free Channel

Initially, the communication channel is assumed to be **error-free**.

This means:

- Frames are not damaged.
- Frames are not lost.
- Bits are not changed during transmission.

Therefore, error detection and retransmission are not considered in the simplest protocol.

4. Receiver is Always Ready

It is assumed that the receiver can always accept incoming frames.

The sender does not need to wait because of receiver processing limitations.

5. Network Layer is Always Ready

It is assumed that the Network Layer at the sender always has data to send.

Therefore, the Data Link Layer always has a packet available for transmission.

6. Frames are Delivered in Order

It is assumed that frames arrive at the receiver in the same order in which they were transmitted.

For example:

```
Sent:      Frame 1 → Frame 2 → Frame 3
Received:  Frame 1 → Frame 2 → Frame 3
```

7. No Frame Duplication

It is assumed that a frame is received only once.

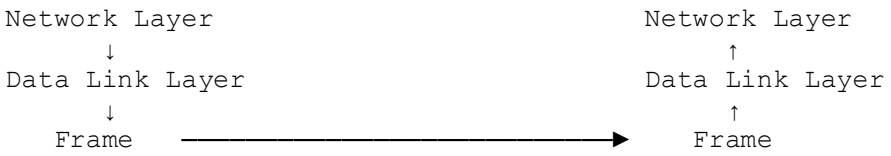
There are no duplicate frames at the receiver.

Basic Transmit and Receive

The basic transmit and receive operation explains how a packet is transferred from the sender's Network Layer to the receiver's Network Layer through the Data Link Layer.

The process can be represented as:

```
  Sender                                    Receiver
```



Basic Transmit Operation

The **transmit operation** is performed by the sender.

Step 1: Receive Packet

The sender's Data Link Layer receives a packet from the Network Layer.

Step 2: Create a Frame

The Data Link Layer adds necessary control information to the packet and creates a **frame**.

```
Packet + Control Information = Frame
```

Step 3: Send the Frame

The frame is passed to the Physical Layer and transmitted through the communication channel.

```
Network Layer
      ↓
Data Link Layer
      ↓
Physical Layer
      ↓
Communication Channel
```

Step 4: Continue Transmission

Under the simplest assumptions, the sender can continue transmitting frames because the receiver is assumed to be ready.

Basic Receive Operation

The **receive operation** is performed by the receiver.

Step 1: Receive Frame

The receiver's Physical Layer receives the transmitted bits from the communication channel.

Step 2: Pass the Frame to Data Link Layer

The received information is passed to the Data Link Layer.

Step 3: Extract the Packet

The Data Link Layer removes the control information and extracts the original packet.

Step 4: Deliver to Network Layer

The extracted packet is passed to the Network Layer.

```
Communication Channel
          ↓
   Physical Layer
          ↓
   Data Link Layer
          ↓
    Network Layer
```

10.Simplex Link Layer Protocols

A **simplex link-layer protocol** is a Data Link Layer protocol in which data flows in **only one direction**.

Simplex means:
Sender → Receiver

The receiver does not send data back to the sender. However, in some protocols, the receiver may send **control information such as acknowledgements** depending on the protocol being studied.

There are three basic simplex protocols:

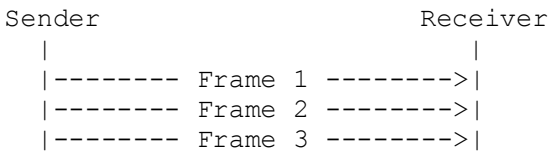
1. Unrestricted Simplex Protocol
2. Simplex Stop-and-Wait Protocol
3. Simplex Protocol for a Noisy Channel

1. Unrestricted Simplex Protocol

This is the simplest protocol.

Assumptions:

- Data flows only from sender to receiver.
- The communication channel is error-free.
- The receiver is always ready to receive.
- The sender always has data to send.
- No flow control is required.
- No acknowledgements are required.

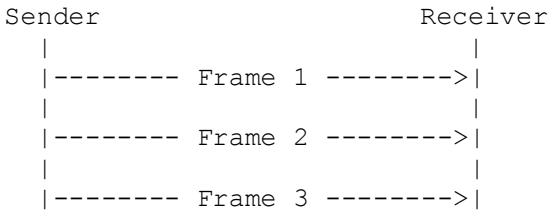


The sender continuously sends frames to the receiver.

2. Simplex Stop-and-Wait Protocol

Here, the receiver may not be able to process frames as quickly as the sender sends them.

Therefore, the sender sends **one frame at a time and waits** before sending the next frame.

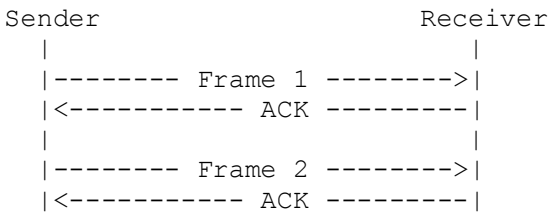


This provides **flow control** and prevents the receiver from being overloaded.

3. Simplex Protocol for a Noisy Channel

This protocol considers the possibility that frames may be **lost or damaged** during transmission.

The sender uses acknowledgements and retransmission to provide reliable communication.



If a frame is lost or damaged, the sender retransmits it.

11. Improving Efficiency in Data Link Layer Protocols

In a simple **Stop-and-Wait protocol**, the sender sends one frame and waits for an acknowledgement before sending the next frame. This is reliable but can be **slow and inefficient**, especially when the communication link has a long delay.

During the time the sender is waiting for the acknowledgement, the communication channel may remain **idle**.

Therefore, to improve efficiency, we need to allow the sender to transmit **multiple frames before waiting for acknowledgements**.

Sliding Window Protocol

The main technique used to improve efficiency is the **Sliding Window Protocol**.

In this method:

- Multiple frames can be transmitted continuously.
- The sender does not have to wait after every frame.
- The receiver acknowledges received frames.
- A **window** keeps track of frames that can be sent.
- As acknowledgements are received, the window moves forward, or **slides**.

Benefits of Improving Efficiency

- **Higher data transmission rate**
- **Better utilization of the communication channel**
- **Less waiting time**
- **Improved network performance**
- **Multiple frames can be transmitted at the same time**
- Better use of links with **long propagation delays**

12. Bi-Directional Transmission

Bi-directional transmission means that data can travel in **both directions** between two devices.

Unlike simplex transmission, where data travels only in one direction, bi-directional communication allows both devices to **send and receive data**.

Simple Example

```
Device A  ⇄  Device B
```

Device A can send data to Device B, and Device B can also send data to Device A.

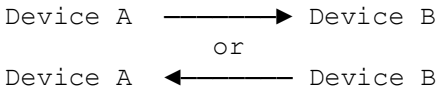
Types of Bi-Directional Transmission

There are two main types:

1. Half-Duplex

In **half-duplex communication**, both devices can transmit data, but **only one device can transmit at a time**.

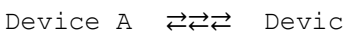
Example:



A common example is a **walkie-talkie**. One person speaks while the other listens. After the first person finishes, the other can speak.

2. Full-Duplex

In **full-duplex communication**, both devices can **send and receive data at the same time**.

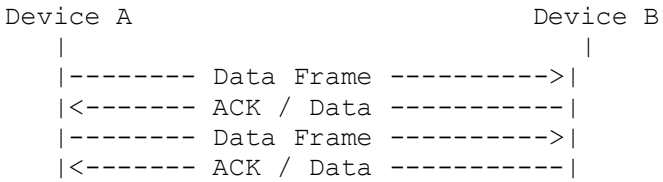


For example, during a telephone call, both people can speak and listen simultaneously.

Bi-Directional Transmission in Data Link Layer

In Data Link Layer protocols, bi-directional communication can improve efficiency because the receiver can send **acknowledgements** while also transmitting its own data.

For example:



This technique can reduce unnecessary waiting and make better use of the communication link.

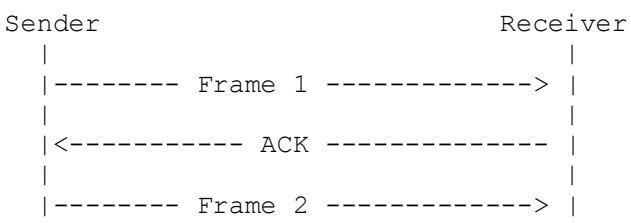
13. Multiple Frames in Flight

Multiple frames in flight means that several frames can be transmitted over the communication link before the sender receives acknowledgements for the earlier frames.

This technique is used to **improve the efficiency of data transmission**.

Need of multiple frames in flight

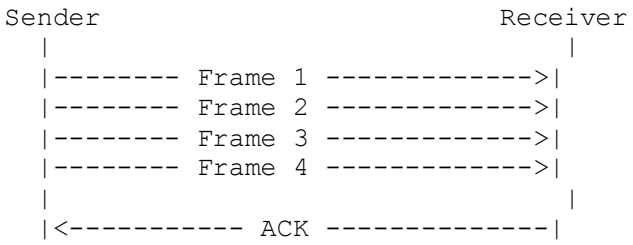
In the **Stop-and-Wait protocol**, the sender sends one frame and waits for an acknowledgement.



The sender remains idle while waiting for the ACK. This wastes the capacity of the communication channel, especially when the channel has a long delay.

Multiple Frames

Instead of waiting after every frame, the sender can send several frames continuously.



Here, **Frames 1, 2, 3, and 4 are in flight** at the same time.

Advantages

1. Improves channel utilization
2. Reduces waiting time.
3. Increases throughput.
4. Allows continuous transmission.
5. Makes better use of high-speed and long-distance links.
6. Reduces the effect of propagation delay.

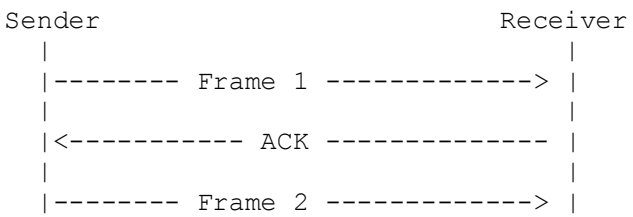
14. Sliding Window Protocol

Sliding Window Protocol is a **Data Link Layer protocol** used for **flow control and error control**. It allows the sender to transmit **multiple frames before receiving acknowledgements**, instead of sending one frame and waiting after each frame.

This improves the **efficiency and speed** of data transmission.

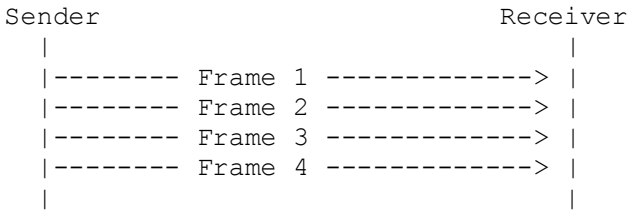
Need of Sliding Window Protocol:

In the **Stop-and-Wait protocol**, the sender sends one frame and waits for an acknowledgement.



The sender remains idle while waiting for the ACK.

Sliding Window solves this problem by allowing several frames to be **in flight** at the same time.



Meaning of "Window"

A **window** is a set of frames that the sender is allowed to transmit without waiting for individual acknowledgements.

For example, if the window size is **4**:

```
[ Frame 1 | Frame 2 | Frame 3 | Frame 4 ]
```

The sender can transmit all four frames.

When an acknowledgement is received, the window **moves forward**, or **slides**.

```
Before:
[ 1 | 2 | 3 | 4 ]

After ACK for Frame 1:
    [ 2 | 3 | 4 | 5 ]
```

This is why it is called the **Sliding Window Protocol**.

Working of Sliding Window Protocol

Step 1: Create Frames

The sender divides the data into smaller units and creates frames.

Step 2: Assign Sequence Numbers

Each frame is given a **sequence number**.

For example:

```
Frame 0
Frame 1
Frame 2
Frame 3
Frame 4
```

Step 3: Define Window Size

The sender has a specific window size.

For example:

Window size = 4

The sender can initially transmit:

```
[ 0 | 1 | 2 | 3 ]
```

Step 4: Transmit Multiple Frames

The sender sends frames 0, 1, 2, and 3 without waiting for individual acknowledgements.

Step 5: Receive Acknowledgement

When the receiver successfully receives frames, it sends acknowledgements.

For example:

```
ACK 0
ACK 1
ACK 2
```

Step 6: Slide the Window

After receiving acknowledgements, the sender moves the window forward.

```
Initial:
[ 0 | 1 | 2 | 3 ]

After ACK 0:
    [ 1 | 2 | 3 | 4 ]

After ACK 1:
        [ 2 | 3 | 4 | 5 ]
```

The sender can now transmit new frames.

Types of Sliding Window Protocols

There are two important types:

1. Go-Back-N ARQ

If one frame is lost or damaged, the sender retransmits the **incorrect frame and all subsequent frames**.

Example:

```
Frame 1 → received
Frame 2 → received
Frame 3 → lost
Frame 4 → received
```

The sender goes back and retransmits:

```
Frame 3 → retransmit
Frame 4 → retransmit
```

2. Selective Repeat ARQ

If one frame is lost or damaged, **only that particular frame is retransmitted**.

Example:

```
Frame 1 → received
Frame 2 → received
Frame 3 → lost
Frame 4 → received
```

Only Frame 3 is retransmitted:

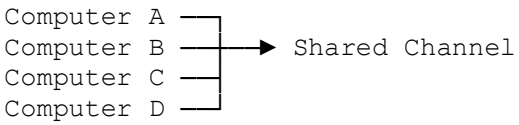
```
Frame 3 → retransmit
```

Therefore, Selective Repeat is generally more efficient than Go-Back-N, especially when errors are frequent.

15. Channel Allocation Problem

In a computer network, many devices may need to use the **same communication channel** to transmit data. The **Channel Allocation Problem** is the problem of deciding **which device can use the shared communication channel and when**.

For example, if many computers are connected to the same network, two or more computers may try to transmit at the same time. This can cause a **collision**, resulting in loss or corruption of data.



Therefore, a suitable channel allocation method is required to use the channel efficiently.

Types of Channel Allocation

Channel allocation is mainly divided into:

1. Static Channel Allocation
2. Dynamic Channel Allocation

1. Static Channel Allocation

In **static channel allocation**, the available communication channel is divided among the users in advance.

Each user is assigned a fixed portion of the channel.

For example, in a network with four users:

```
Channel
| User A | User B | User C | User D |
```

Each user receives a fixed portion of the available bandwidth.

Example: Frequency Division

Suppose a communication channel has a bandwidth of 100 MHz. It can be divided among four users:

```
User A → 25 MHz
User B → 25 MHz
User C → 25 MHz
User D → 25 MHz
```

Each user gets a fixed portion.

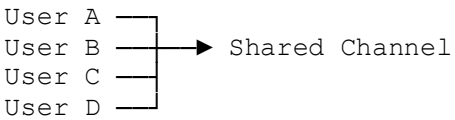
Advantages

- Simple to implement.
- No collisions between users.
- Predictable allocation.
- Suitable when users continuously have data to transmit.

2. Dynamic Channel Allocation

In **dynamic channel allocation**, the channel is shared dynamically according to the current requirements of the users.

A device gets access to the channel when it has data to transmit.



If User A has no data to send, another user can use the available channel capacity.

Advantages

- Better utilization of the channel.
- Suitable for bursty traffic.
- Flexible allocation.
- Reduces wastage of bandwidth.

Important Channel Allocation Methods

Several techniques are used to solve the channel allocation problem.

1. Frequency Division Multiple Access (FDMA)

The available frequency band is divided into different frequency ranges, and each user gets a specific frequency.

```
Frequency
| A | B | C | D |
```

2. Time Division Multiple Access (TDMA)

The channel is divided into **time slots**, and each user gets a specific time slot.

```
Time
| A | B | C | D | A | B | C | D |
```

3. Code Division Multiple Access (CDMA)

All users can use the same frequency at the same time, but each user is assigned a different code.

4. Random Access

In random-access methods, a device transmits when it has data.

Examples include:

- **ALOHA**
- **Slotted ALOHA**
- **CSMA**
- **CSMA/CD**
- **CSMA/CA**

5. Controlled Access

In controlled-access methods, devices coordinate their access to the shared channel.

Examples include:

- **Reservation**
- **Polling**
- **Token passing**

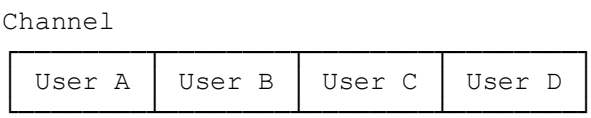
16. Static Channel Allocation

Static Channel Allocation is a method in which the available communication channel is **divided among users in advance**. Each user is permanently assigned a specific portion of the channel, such as a particular **frequency band, time slot, or code**.

The allocation does not change according to the user's current data requirements.

Simple Example

Suppose four computers share a communication channel:



Each user gets a fixed portion of the channel.

Working of Static channel allocation

- The total communication channel is divided into smaller portions.
- Each portion is assigned to a particular user.
- The assignment is made before communication begins.
- A user can use only its assigned portion.
- The allocation generally remains fixed even if the user has no data to transmit.

For example, if a channel has **100 Mbps** capacity and four users are assigned equally:

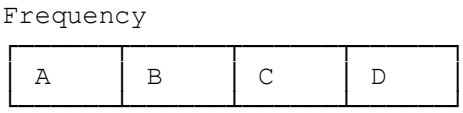
```
User A → 25 Mbps
User B → 25 Mbps
User C → 25 Mbps
User D → 25 Mbps
```

Even if User D is not transmitting, its 25 Mbps portion may remain unused.

Common Static Allocation Techniques

1. Frequency Division Multiple Access (FDMA)

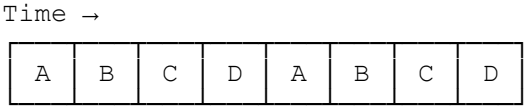
The channel is divided into different **frequency bands**.



Each user continuously gets a specific frequency range.

2. Time Division Multiple Access (TDMA)

The channel is divided into **fixed time slots**.



Each user gets a particular time slot.

3. Code Division Multiple Access (CDMA)

Each user is assigned a **unique code**. Multiple users can share the same frequency band while being separated using their codes.

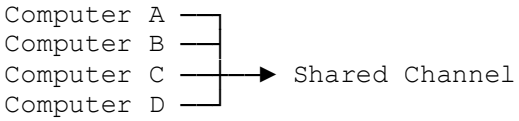
17. Dynamic Channel Allocation

ynamic Channel Allocation is a method in which a shared communication channel is **allocated to users according to their current requirements**.

Unlike static channel allocation, where each user gets a fixed portion of the channel, dynamic allocation allows users to **access the channel when they have data to transmit**.

Simple Example

Suppose four computers share one communication channel:



If only Computer A has data to send, it can use the channel. When A finishes, another computer can use it.

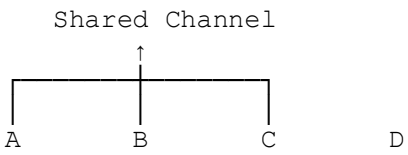
Working of Dynamic Channel Allocation

The basic operation is:

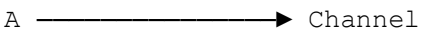
1. Several devices share a common communication channel.
2. A device requests or attempts to access the channel when it has data.
3. The channel allocation mechanism determines whether the device can transmit.
4. The device transmits its data.
5. After transmission, the channel becomes available to other devices.
6. If multiple devices attempt to transmit simultaneously, a **collision** may occur in some dynamic-access methods.
7. The protocol handles the collision or schedules another transmission.

Example

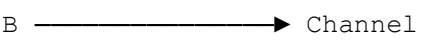
Consider four computers:



Initially, only A has data:



Later, B has data:



The channel is therefore **allocated according to demand**, rather than permanently assigned to one computer.

Important Dynamic Channel Allocation Methods

1. Random Access

In random-access methods, a station attempts to access the channel whenever it has data.

Examples:

- **ALOHA**
- **Slotted ALOHA**
- **CSMA**
- **CSMA/CD**
- **CSMA/CA**

For example, in CSMA, a station first checks whether the channel is free before transmitting.

2. Controlled Access

In controlled-access methods, devices coordinate with each other to determine who can use the channel.

Examples:

- **Reservation**
- **Polling**
- **Token Passing**

3. Channelization

The available channel is shared among users using techniques such as:

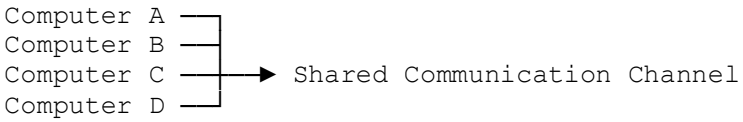
- **FDMA**
- **TDMA**
- **CDMA**

Depending on the system, these resources can be assigned according to current requirements.

18. Multiple Access Protocols

In a computer network, many devices may need to use the **same communication channel** to send data. If two or more devices transmit at the same time, their signals may interfere with each other, causing a **collision**.

Multiple Access Protocols are rules that determine **how multiple devices share and access a common communication channel**.



The main goal is to provide **efficient and fair access** to the shared channel while reducing collisions.

Need for Multiple Access Protocols

Multiple access protocols are required to:

1. Control access to a shared channel.
2. Reduce or avoid collisions.
3. Improve channel utilization.
4. Provide fair access to different devices.
5. Increase network performance.
6. Reduce data loss and retransmissions.

Types of Multiple Access Protocols

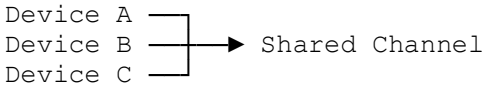
Multiple Access Protocols are broadly classified into:

1. Random Access Protocols
2. Controlled Access Protocols
3. Channelization Protocols

1. Random Access Protocols

In **Random Access Protocols**, any device can attempt to transmit data whenever it has data to send.

There is **no fixed schedule** for transmission.



If two devices transmit at the same time, a **collision** may occur.

Important Random Access Protocols

- **ALOHA**
- **Slotted ALOHA**
- **CSMA**
- **CSMA/CD**

- **CSMA/CA**

1.ALOHA

In ALOHA, a device transmits whenever it has data.

If a collision occurs, the device waits for a random period and retransmits.

2.CSMA

CSMA (Carrier Sense Multiple Access) requires a device to **listen to the channel before transmitting**.

- If the channel is free → transmit.
- If the channel is busy → wait.

3.CSMA/CD

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) detects a collision while transmission is taking place.

It was traditionally associated with shared, half-duplex Ethernet.

4.CSMA/CA

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) attempts to avoid collisions before transmission and is commonly associated with wireless LANs.

2. Controlled Access Protocols

In **Controlled Access Protocols**, devices coordinate their access to the channel.

Instead of allowing every device to transmit whenever it wants, the protocol determines **which device gets permission to transmit**.

Main Controlled Access Methods

1. **Reservation**
2. **Polling**
3. **Token Passing**

Reservation

Devices reserve the channel before transmitting.

```
Reservation → Transmission
```

This helps avoid collisions because devices transmit according to their reservations.

Polling

A central device controls access to the channel.

The central controller asks each device whether it has data to send.

```
Controller → Device A
Controller → Device B
Controller → Device C
```

The selected device gets permission to transmit.

Token Passing

A special control frame called a **token** is passed between devices.

Only the device holding the token is allowed to transmit.

```
A → B → C → D → A
```

This prevents collisions because only one device has permission to transmit at a time.

3. Channelization Protocols

In **Channelization Protocols**, the available communication channel is divided among multiple users.

The users are separated based on **frequency, time, or code**.

Main Channelization Methods

1. **FDMA**
2. **TDMA**
3. **CDMA**

FDMA

Frequency Division Multiple Access divides the available bandwidth into different frequency bands.

```
Frequency
| A | B | C | D |
```

Each user gets a separate frequency band.

TDMA

Time Division Multiple Access divides the channel into time slots.

```
Time →
| A | B | C | D | A | B | C | D |
```

Each user gets a specific time slot.

CDMA

Code Division Multiple Access allows multiple users to use the same frequency at the same time, but each user is assigned a unique code.

19. ALOHA Protocol

ALOHA is a **Random Access Protocol** used at the **Data Link Layer** to allow multiple devices to share a common communication channel.

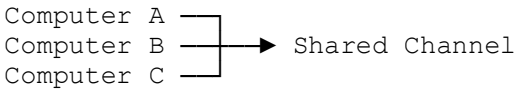
The basic idea is very simple:

A device transmits whenever it has data to send.

There is no fixed time slot or central controller that decides which device can transmit.

Working of ALOHA

Suppose several computers share one communication channel:



The operation is:

1. A device gets a frame to transmit.
2. It **transmits immediately**.
3. If no other device transmits at the same time, the frame is successfully received.
4. If two or more devices transmit at the same time, a **collision** occurs.
5. The devices wait for a **random period of time**.
6. They retransmit the frames

Types of ALOHA

There are two main types:

1. Pure ALOHA

In **Pure ALOHA**, a device can transmit **at any time** when it has data.

2. Slotted ALOHA

In **Slotted ALOHA**, time is divided into fixed **time slots**.

A device can start transmitting only at the **beginning of a time slot**.

20. Carrier Sense Multiple Access (CSMA) Protocol

CSMA stands for **Carrier Sense Multiple Access**. It is a **Random Access Protocol** used at the Data Link Layer to control how multiple devices share a common communication channel.

The main idea of CSMA is:

Listen before you transmit.

Before sending data, a device first checks whether the communication channel is free or busy.

Need of CSMA

If several devices share the same channel and transmit at the same time, a **collision** may occur.

CSMA tries to reduce collisions by making a device **sense (listen to) the channel before transmitting**.



Working of CSMA

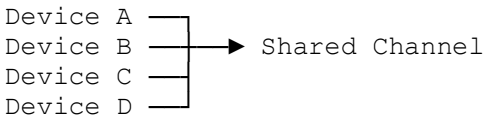
The basic operation is:

1. A device has a frame to transmit.
2. It **senses the channel**.
3. If the channel is **idle**, it transmits the frame.
4. If the channel is **busy**, it waits according to the CSMA method being used.
5. After the channel becomes available, the device transmits.

21. Collision-Free Protocols

In a shared communication network, many devices may need to use the **same communication channel**. If two or more devices transmit at the same time, a **collision** can occur.

Collision-free protocols are protocols designed to **prevent collisions by controlling which device is allowed to transmit and when**.



Instead of allowing devices to compete randomly for the channel, collision-free protocols organize access to the channel.

Need of Collision Free Protocol

They are used to:

1. Prevent data collisions.
2. Improve channel utilization.
3. Provide fair access to all devices.
4. Reduce retransmissions.
5. Improve network performance.
6. Provide predictable access to the communication channel.

Important Collision-Free Protocols

The major collision-free protocols include:

1. **Bit-Map Protocol**
2. **Binary Countdown Protocol**
3. **Token Passing**
4. **Reservation Protocol**

1. Bit-Map Protocol

The **Bit-Map Protocol** uses a small reservation period before the actual data transmission.

Each device has a fixed position in a **bit map**.

Suppose there are four devices:

```
A   B   C   D
0   0   1   0
```

Here, `1` indicates that the corresponding device wants to transmit.

In this example, only **Device C** wants to transmit.

Working

1. Each device gets a chance to indicate whether it wants to transmit.
2. A device places 1 in its assigned position if it has data.
3. Devices with 0 do not transmit.
4. After the reservation period, devices transmit in an agreed order.
5. Since the transmission order is known, collisions are avoided.

2. Binary Countdown Protocol

The **Binary Countdown Protocol** is another collision-free protocol.

In this method, each device has a unique **binary address**.

When multiple devices want to transmit, they participate in a competition based on their binary addresses.

For example:

```
Device A → 001
Device B → 010
Device C → 101
```

The devices compare their addresses bit by bit. The device with the **highest priority address** wins the right to transmit.

For example:

```
A = 001
B = 010
C = 101
```

Since `101` is the highest value, **Device C wins**.

The winning device transmits its frame, while the other devices wait.

3. Token Passing

In **Token Passing**, a special control frame called a **token** circulates among the devices.

Only the device holding the token is allowed to transmit.

```
A → B → C → D → A
```

Suppose Device B has the token:

```
A       B       C       D
        ↓
      TOKEN
```

B can transmit its data.

After finishing, B passes the token to C.

```
A → B → C → D → A
        TOKEN
```

Because only one device has the token at a time, **collisions are avoided**.

4. Reservation Protocol

In a **Reservation Protocol**, devices first reserve the channel before transmitting their data.

```
Reservation Phase
        ↓
Transmission Phase
```

Devices that want to transmit indicate their requirement during the reservation phase.

Then they transmit according to the reservation order.

22. Limited Contention Protocols

Limited Contention Protocols are channel access protocols that combine the advantages of **random access** and **collision-free access**.

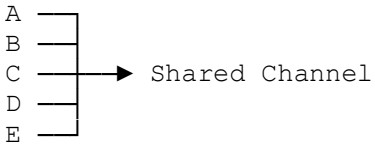
In a random-access protocol, any device may try to transmit, which can cause collisions. In a collision-free protocol, access is carefully controlled, which reduces collisions but may introduce more overhead.

Limited contention tries to achieve a balance between these two approaches.

Main idea: Allow a limited number of devices to compete for the channel so that the probability of collision is reduced.

Need for Limited Contention

Consider many devices sharing one channel:



If all devices are allowed to compete at the same time, many collisions may occur.

Limited contention divides the devices into **groups** and controls which group can compete for the channel.

Working

The basic operation is:

1. Devices are divided into different groups.
2. Only a selected group is allowed to compete for the channel during a particular period.
3. Devices in that group use a contention method to obtain access.
4. Once a device gets access, it transmits its frame.
5. The next group is then considered.

```
Group 1 → Contention → Transmission
Group 2 → Contention → Transmission
Group 3 → Contention → Transmission
```

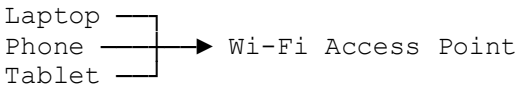
This limits the number of devices competing at one time.

23. Wireless LAN Protocol

A **Wireless LAN (WLAN) protocol** is a set of rules used to control communication between devices over a **wireless local area network**.

The most common WLAN technology is **Wi-Fi**, which is based on the **IEEE 802.11** family of standards.

Unlike wired Ethernet, wireless devices share a radio channel, so controlling access to the channel is important.



Need of WLAN Protocol

In a wireless network, many devices may use the same radio channel. If two devices transmit simultaneously, their signals can interfere.

Therefore, WLAN protocols are required to:

- Control access to the wireless channel.
- Reduce collisions.
- Provide reliable data transmission.
- Manage communication between wireless stations.
- Improve channel utilization.

CSMA/CA

The main channel-access mechanism used by traditional IEEE 802.11 WLANs is **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)**.

The basic idea is:

Listen before transmitting and try to avoid collisions.

Working of CSMA/CA

1. A wireless station has a frame to transmit.
2. It **checks whether the wireless channel is busy or idle**.
3. If the channel is busy, it waits.
4. If the channel is idle for the required period, it uses a **random backoff** before transmitting.
5. The station transmits when its backoff reaches zero.

6. The receiver sends an **ACK** if the frame is received successfully.
7. If the sender does not receive the ACK, it assumes that the transmission was unsuccessful and may retransmit.

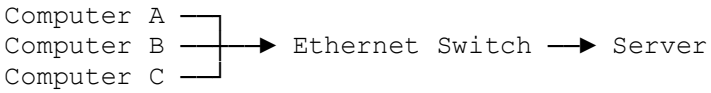
24. Ethernet

Ethernet is a widely used **Local Area Network (LAN) technology** for connecting computers, servers, switches, printers, and other network devices.

Ethernet defines how devices **format, transmit, receive, and identify data** over a network.

Ethernet is standardized by **IEEE 802.3**.

Simple Example



Here, Ethernet provides the rules for communication between the connected devices.

Main Features of Ethernet

1. Used mainly in **Local Area Networks (LANs)**.
2. Uses **frames** to transmit data.
3. Uses **MAC addresses** to identify devices.
4. Defined by the **IEEE 802.3** standards.
5. Supports different data rates.
6. Can use copper cables or fiber-optic cables.
7. Modern Ethernet commonly uses **switches**.

25. Classic Ethernet Physical Layer

lassic Ethernet refers to the original Ethernet technologies that were developed for **shared-medium local area networks**. The **Physical Layer** is responsible for transmitting raw bits over the physical communication medium.

Classic Ethernet is standardized under **IEEE 802.3**.

The Physical Layer defines:

- The transmission medium
- Electrical or optical signals
- Data transmission rate
- Connectors and cables
- Maximum cable length
- How bits are represented as signals

Main Classic Ethernet Standards

Classic Ethernet originally supported a data rate of **10 Mbps**.

The important physical-layer versions are:

Ethernet Type	Cable	Speed	Maximum Segment Length
10BASE5	Thick coaxial cable	10 Mbps	500 m
10BASE2	Thin coaxial cable	10 Mbps	200 m
10BASE-T	Twisted-pair cable	10 Mbps	100 m

Meaning of 10BASE5

The name can be understood as:

- **10** → 10 Mbps
- **BASE** → Baseband transmission
- **5** → approximately 500 meters maximum segment length

Similarly:

10BASE2

- **10** → 10 Mbps
- **BASE** → Baseband
- **2** → approximately 200 meters

26. Classic Ethernet MAC Sublayer Protocol

The **MAC (Media Access Control) sublayer** is a part of the **Data Link Layer**. In Classic Ethernet, the MAC sublayer controls **how multiple devices share a common communication medium**.

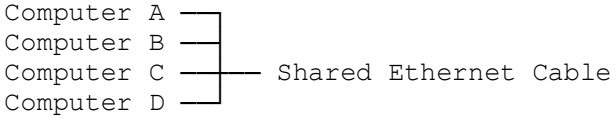
Classic Ethernet originally used a **shared channel**, where many computers were connected to the same physical medium. Therefore, a method was needed to decide when a computer could transmit.

The main MAC protocol used in Classic Ethernet was:

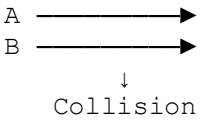
CSMA/CD — Carrier Sense Multiple Access with Collision Detection

Need for CSMA/CD

Consider several computers sharing one Ethernet cable:



If A and B transmit at the same time, their signals interfere.



CSMA/CD was designed to handle this situation.

CSMA/CD

CSMA/CD stands for:

- **C** – Carrier
- **S** – Sense
- **M** – Multiple
- **A** – Access
- **C** – Collision
- **D** – Detection

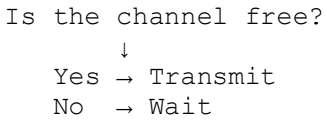
The basic idea is:

Listen before transmitting, and detect a collision if one occurs.

Working of CSMA/CD

Step 1: Carrier Sense

Before transmitting, a station(computers) first **listens to the Ethernet channel**.



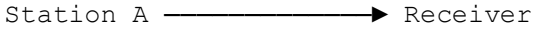
Step 2: Multiple Access

The same communication medium is shared by **multiple stations**.

Any station that has data to send can attempt to access the channel.

Step 3: Transmit

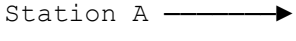
If the channel is idle, the station starts transmitting its Ethernet frame.

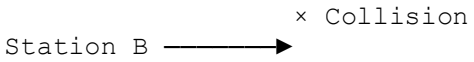


Step 4: Collision Detection

While transmitting, the station continues monitoring the channel.

If another station also transmits, a collision occurs.





The transmitting stations detect the collision.

Step 5: Send Jam Signal

When a collision is detected, the stations stop normal transmission and send a **jam signal** so that all stations become aware that a collision has occurred.

Step 6: Backoff

After the collision, each station waits for a **random amount of time** before trying again.

Ethernet uses **binary exponential backoff** to determine this waiting period.

Step 7: Retransmission

After the waiting period, the station senses the channel again.

If the channel is free, it retransmits the frame.

27. Fast Ethernet, Gigabit Ethernet, 10-Gigabit Ethernet,40-Gigabit Ethernet,& 100- Gigabit Ethernet.

Ethernet has evolved over time to provide **higher data transmission speeds**. The original Classic Ethernet operated at **10 Mbps**. As network requirements increased, faster versions of Ethernet were developed.

The major Ethernet technologies are:

- **Fast Ethernet** – 100 Mbps
- **Gigabit Ethernet** – 1 Gbps
- **10-Gigabit Ethernet** – 10 Gbps
- **40-Gigabit Ethernet** – 40 Gbps
- **100-Gigabit Ethernet** – 100 Gbps

All are part of the **IEEE 802.3 Ethernet family**.

1. Fast Ethernet

Fast Ethernet provides a data rate of **100 Mbps**, which is ten times faster than Classic Ethernet.

It is commonly associated with **IEEE 802.3u**.

Features

- Speed: **100 Mbps**

- Uses copper or fiber
- Supports full-duplex operation
- Commonly used in older LAN installations

2. Gigabit Ethernet

Gigabit Ethernet provides a data rate of **1 Gbps (1000 Mbps)**.

It is standardized through several IEEE 802.3 specifications, including **802.3ab** for 1000BASE-T and **802.3z** for several fiber implementations.

Features

- Speed: **1 Gbps**
- Supports copper and fiber
- Commonly used in LANs
- Supports full-duplex operation

3. 10-Gigabit Ethernet

10-Gigabit Ethernet provides a data rate of **10 Gbps**.

It is designed for high-speed communication in **enterprise networks, data centers, and backbone networks**.

Features

- Speed: **10 Gbps**
- Mainly used for high-speed network connections
- Supports fiber and copper variants
- Commonly used in data centers and network backbones

4. 40-Gigabit Ethernet

40-Gigabit Ethernet provides a data rate of **40 Gbps**.

It was developed to meet the increasing bandwidth requirements of **data centers, high-performance computing, and network backbones**.

Features

- Speed: **40 Gbps**
- Mainly used for high-capacity links
- Common in data-center and backbone environments
- Primarily uses optical technologies for longer-distance applications

5. 100-Gigabit Ethernet

100-Gigabit Ethernet provides a data rate of **100 Gbps**.

It was developed for very high-bandwidth applications such as **large data centers, telecommunications networks, cloud infrastructure, and network backbones**.

Features

- Speed: **100 Gbps**
- Very high bandwidth
- Widely relevant to data centers and backbone networks
- Primarily uses high-speed optical or specialized electrical interfaces depending on the implementation